

CHENGDE, China

WMO: 544230

Lat: 40.9726N

Lon: 117.9247E

Elev: 424

StdP: 96.34

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-16.4	-28.7	0.3	-12.8	-26.4	0.4	-11.4	7.6	-6.7	6.7	-7.7	0.9	340	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.4	20.7	31.9	20.7	30.5	20.4	24.7	29.1	23.8	28.2	23.0	27.3	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	19.4	27.3	22.5	18.2	26.3	21.6	17.2	25.5	78.2	29.2	74.1	28.3	70.7	27.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.3	5.9	4.8	DB	-20.9	36.3	2.5	2.0	-22.7	37.8	-24.1	38.9	-25.5	40.0	-27.3	41.4
(5)				WB	-21.4	26.0	2.4	1.1	-23.1	26.7	-24.5	27.4	-25.8	28.0	-27.5	28.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-8.4	-3.9	3.5	11.8	17.9	22.2	24.7	23.2	17.7	9.9	0.7	-6.8	(6)
(7)		DBStd	12.11	3.33	4.05	4.51	4.26	3.49	2.84	2.25	2.45	3.33	3.82	4.31	3.38	(7)
(8)		HDD10.0	2047	571	390	206	29	1	0	0	1	48	279	522		(8)
(9)		HDD18.3	3786	829	623	458	200	49	4	0	1	49	261	529	781	(9)
(10)		CDD10.0	1842	0	0	6	82	246	366	456	411	232	46	0	0	(10)
(11)		CDD18.3	539	0	0	0	3	36	120	197	153	31	1	0	0	(11)
(12)		CDH23.3	5119	0	0	2	75	543	1177	1756	1239	318	10	0	0	(12)
(13)		CDH26.7	1838	0	0	0	11	173	462	703	419	69	1	0	0	(13)
(14)	Wind	WSAvg	1.5	1.4	1.6	2.0	2.2	2.0	1.5	1.3	1.1	1.1	1.3	1.3	1.3	(14)
(15)	Precipitation	PrecAvg	521	2	5	8	20	44	83	148	130	52	19	6	2	(15)
(16)		PrecMax	726	27	22	22	70	95	173	293	275	99	56	22	13	(16)
(17)		PrecMin	328	0	0	0	1	12	25	40	27	3	0	0	0	(17)
(18)		PrecStd	114	5	6	6	16	22	39	50	70	27	14	6	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	10.6	21.5	28.2	33.7	35.3	36.1	34.0	30.9	24.4	14.9	4.8	(19)
(20)		MCWB	-1.7	2.9	9.3	14.3	17.7	18.7	22.0	21.2	19.8	14.1	6.6	-0.6		(20)
(21)		2%	DB	1.5	7.9	17.7	25.6	30.6	33.0	33.5	31.9	28.6	21.7	12.4	2.7	(21)
(22)		MCWB	-3.1	1.2	7.4	12.4	16.1	19.0	22.3	22.0	18.2	12.6	5.7	-1.7		(22)
(23)		5%	DB	-0.3	5.9	14.8	23.2	28.4	31.2	31.9	30.5	26.8	19.4	10.3	0.9	(23)
(24)		MCWB	-4.3	0.2	5.7	11.4	15.2	18.6	22.2	21.5	17.4	11.5	4.6	-2.9		(24)
(25)		10%	DB	-1.9	3.9	12.1	20.9	26.4	29.4	30.4	29.1	25.0	17.3	8.0	-0.8	(25)
(26)		MCWB	-5.3	-1.2	4.0	10.2	14.3	18.4	22.0	21.3	16.7	10.2	3.1	-4.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	4.1	10.4	16.4	19.8	22.9	26.4	26.0	22.0	16.3	8.3	0.1	(27)
(28)		MCDWB	3.0	9.4	18.8	24.6	28.4	29.4	31.0	30.1	27.5	21.8	12.6	3.8		(28)
(29)		2%	WB	-2.8	1.8	8.0	14.1	18.1	21.6	25.2	24.6	20.4	14.4	6.5	-1.4	(29)
(30)		MCDWB	1.0	7.3	16.5	22.8	27.1	27.6	29.7	28.7	25.5	19.0	11.0	2.4		(30)
(31)		5%	WB	-4.0	0.3	6.2	12.5	16.9	20.8	24.3	23.6	19.2	13.0	5.2	-2.7	(31)
(32)		MCDWB	-0.6	5.5	13.7	20.5	24.9	26.9	28.7	27.8	24.2	17.2	9.4	0.4		(32)
(33)		10%	WB	-5.2	-1.0	4.5	11.1	15.9	20.0	23.5	22.7	18.0	11.5	3.6	-3.9	(33)
(34)		MCDWB	-2.0	3.7	11.5	19.1	23.2	25.7	27.9	26.7	22.6	15.8	7.8	-1.0		(34)
(35)	Mean Daily Temperature Range	MDBR	11.8	12.5	13.0	13.8	14.2	12.4	10.4	10.8	12.7	12.8	11.4	11.1		(35)
(36)		5% DB	MCDBR	13.7	15.1	16.9	17.6	18.3	15.9	13.4	13.0	15.4	15.3	14.0	12.8	(36)
(37)		MCWBR	10.0	10.1	9.4	8.0	7.0	5.0	4.4	4.6	6.0	7.7	8.4	9.1		(37)
(38)		5% WB	MCDBR	13.1	14.4	15.8	14.9	14.7	11.5	9.7	9.5	11.7	12.7	12.8	12.2	(38)
(39)		MCWBR	9.8	9.7	9.0	7.3	6.2	4.6	4.1	4.2	5.4	7.1	8.1	8.8		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.428	0.470	0.550	0.600	0.719	0.703	0.637	0.577	0.521	0.394	0.353		(40)
(41)		taud	2.166	1.953	1.892	1.763	1.687	1.530	1.602	1.707	1.770	1.823	2.083	2.161		(41)
(42)		Ebn at Noon	779	748	766	734	709	628	632	659	663	641	707	730		(42)
(43)		Edh at Noon	103	149	177	216	239	280	258	226	199	168	111	96		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.57	3.46	4.70	5.45	6.10	5.44	5.07	5.04	4.43	3.46	2.59	2.24		(44)
(45)		RadStd	0.16	0.28	0.35	0.33	0.41	0.51	0.39	0.32	0.28	0.27	0.27	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page